

Pipe Thickness Loss Analysis

Market pipe dataset • 1,000 records • 11 variables

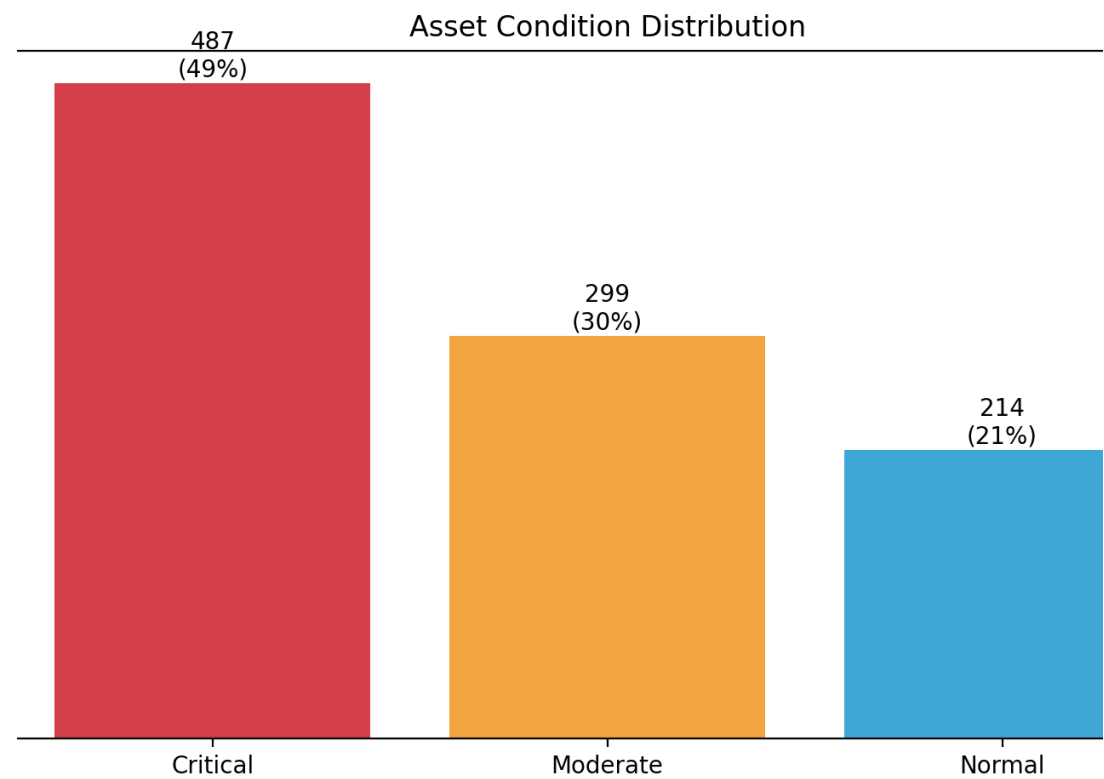
Key headline

Nearly half the pipes are classified Critical, and thickness loss dominates the condition signal.

487 Critical assets

Data quality & profile

The dataset is clean; no missing values or duplicate rows were detected.



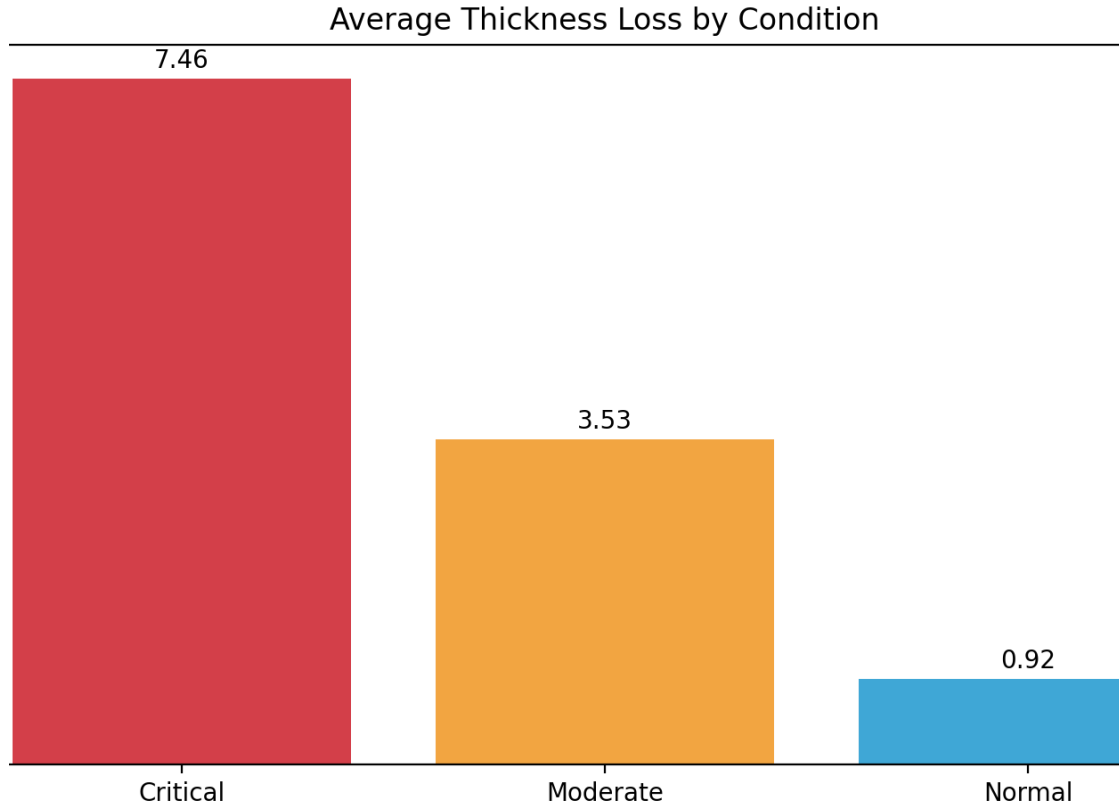
Dataset profile

Records: 1,000
Variables: 11
Materials: 5
Grades: 5
Pipe sizes: 13
Missing values: 0
Duplicate rows: 0

Condition split: 49% Critical, 30% Moderate, 21% Normal

Condition is strongly separated by thickness loss

Average thickness loss rises sharply from Normal to Moderate to Critical.



Interpretation

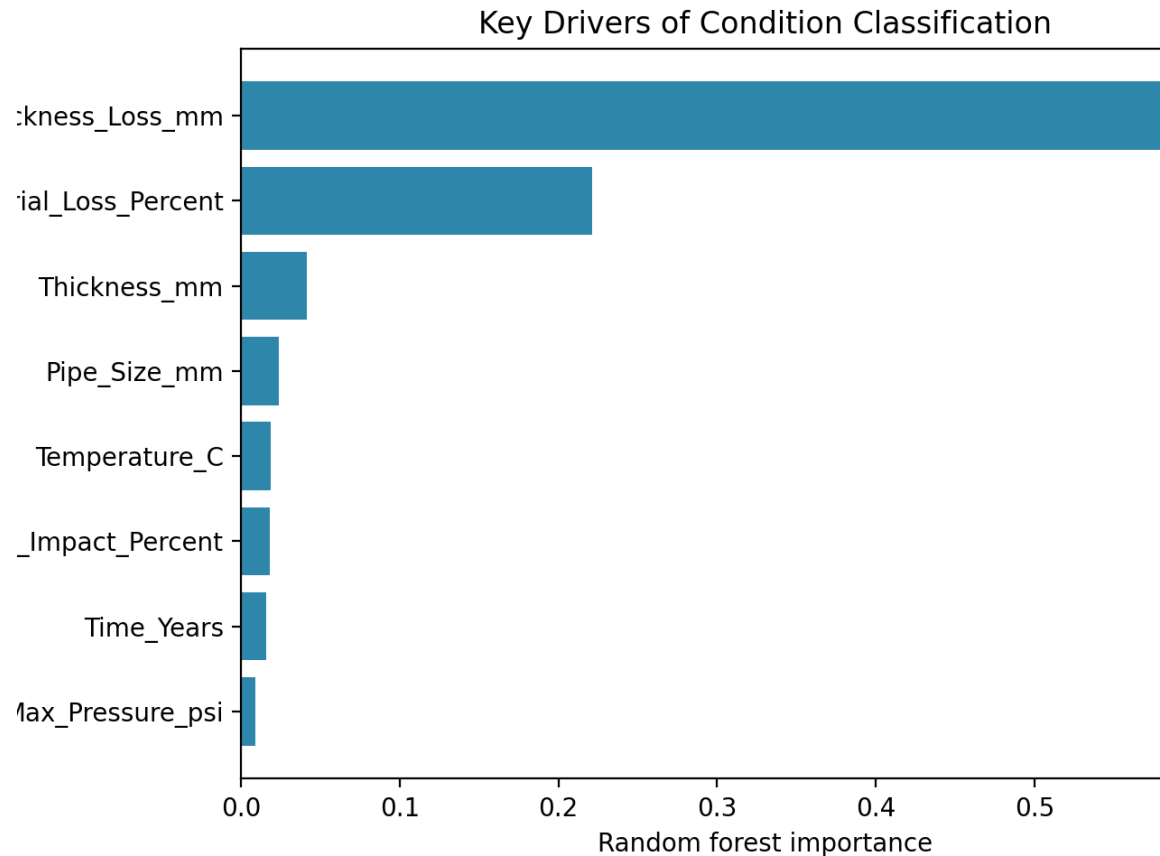
Critical assets average 7.46 mm of thickness loss — more than 8× the Normal average of 0.92 mm.

The practical triage threshold is clear: assets above ~5 mm loss are predominantly Critical in this dataset.

Priority cue: focus inspection capacity on high thickness-loss assets first.

Primary drivers of condition classification

A random-forest classifier confirms that direct loss metrics explain the condition label.



Model readout

5-fold cross-validated accuracy: 99.8%

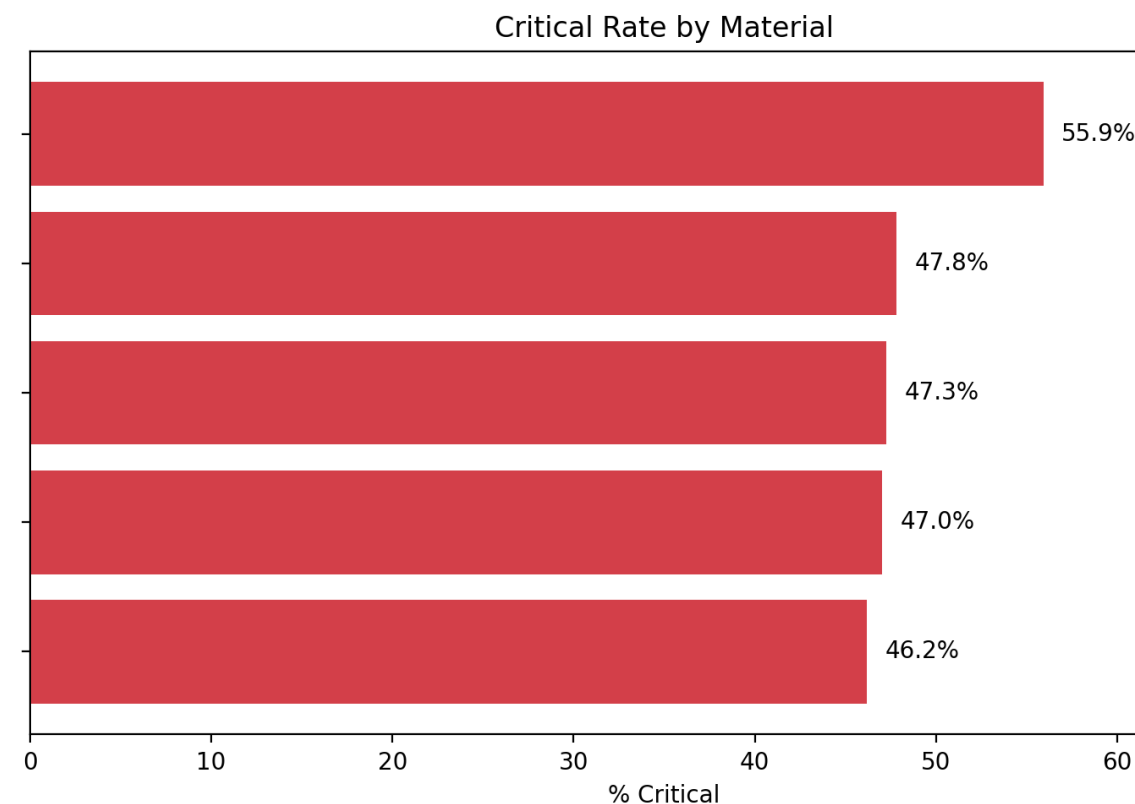
Top drivers:

1. Thickness_Loss_mm
2. Material_Loss_Percent
3. Thickness_mm
4. Pipe_Size_mm

Pressure, temperature, corrosion impact, and age showed weak standalone correlations with thickness loss.

Material and grade risk concentration

PVC has the highest Critical rate; API 5L X65 and X52 rank highest among grades.



Top segments

By material:

PVC: 55.9% Critical

HDPE: 47.8% Critical

Stainless Steel: 47.3% Critical

By grade:

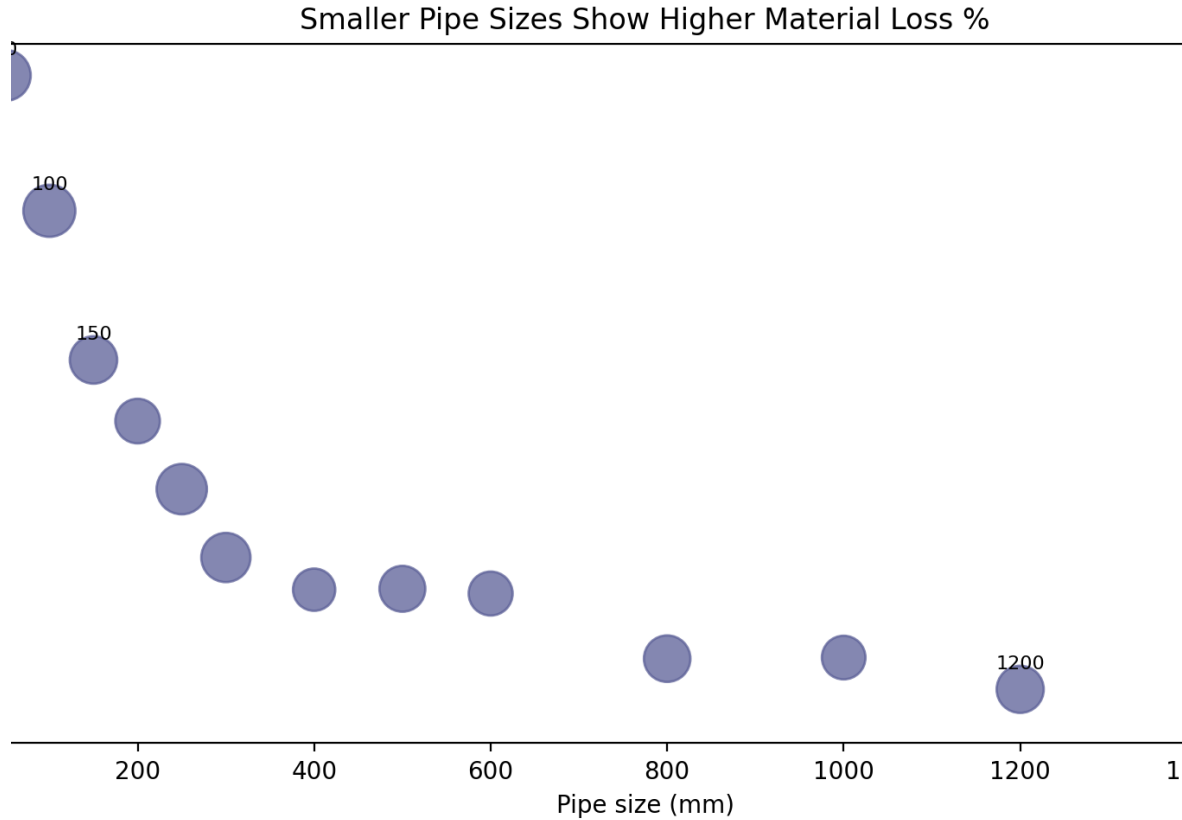
API 5L X65: 53.0% Critical

API 5L X52: 52.9% Critical

Use these segments for risk-based sampling, but avoid over-weighting material alone: direct loss metrics are much stronger indicators.

Pipe size lens: small pipes show higher material loss percentage

Percentage loss is inflated for smaller baseline wall thickness, even when mm loss is similar.



Insight

Average material-loss percentage is highest at 50 mm and 100 mm pipe sizes.

This is because smaller pipes have thinner starting walls; the same millimeter loss translates into a larger percentage loss.

Recommendation: use both absolute mm loss and percentage loss in triage rules.

Recommended action plan — prioritized by risk and consequence

Use condition severity, consequence proxies, and deterioration velocity to sequence work.

1

Tier 1: intervene on high-consequence Critical assets

Inspect the 269 Critical records that also operate at $\geq 1,500$ psi or have diameter ≥ 800 mm. Treat this as the immediate integrity queue.

2

Tier 2: address remaining Critical and near-Critical assets

Schedule the remaining 218 Critical records next, then review 108 Moderate records already at ≥ 4.5 mm loss or $\geq 50\%$ material loss.

3

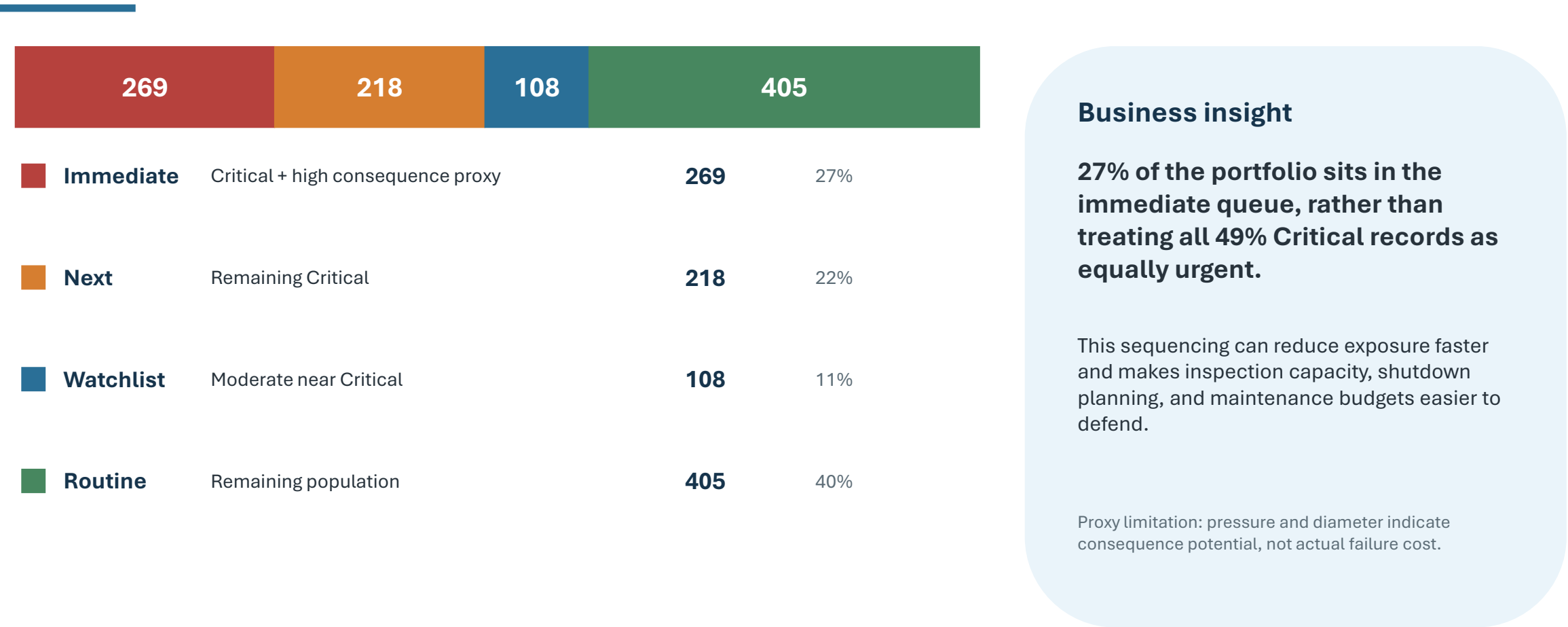
Tier 3: institutionalize risk-based integrity management

Add location, service fluid, coating, inspection dates/method, repair history, shutdown cost, and minimum allowable wall thickness to quantify probability and consequence of failure.

Bottom line: operational risk is concentrated in high-loss assets, not broadly explained by environment variables in this dataset.

A four-tier intervention queue concentrates effort where it matters

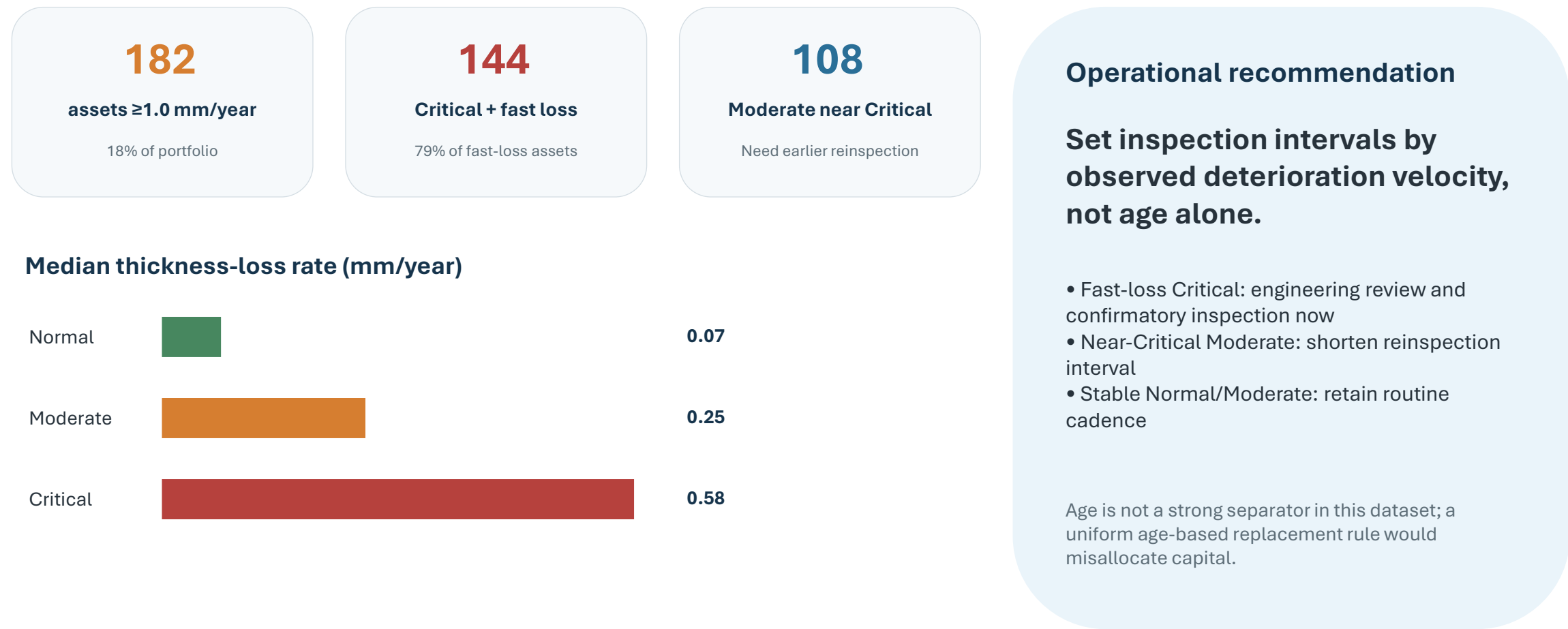
Severity is combined with pressure and diameter as practical consequence proxies.



Decision: reserve emergency/near-term capacity for 269 assets; use planned outages for the next 218 Critical records.

Deterioration velocity identifies assets that may outrun the inspection cycle

Loss per service year is a screening indicator; validate it against actual inspection intervals.



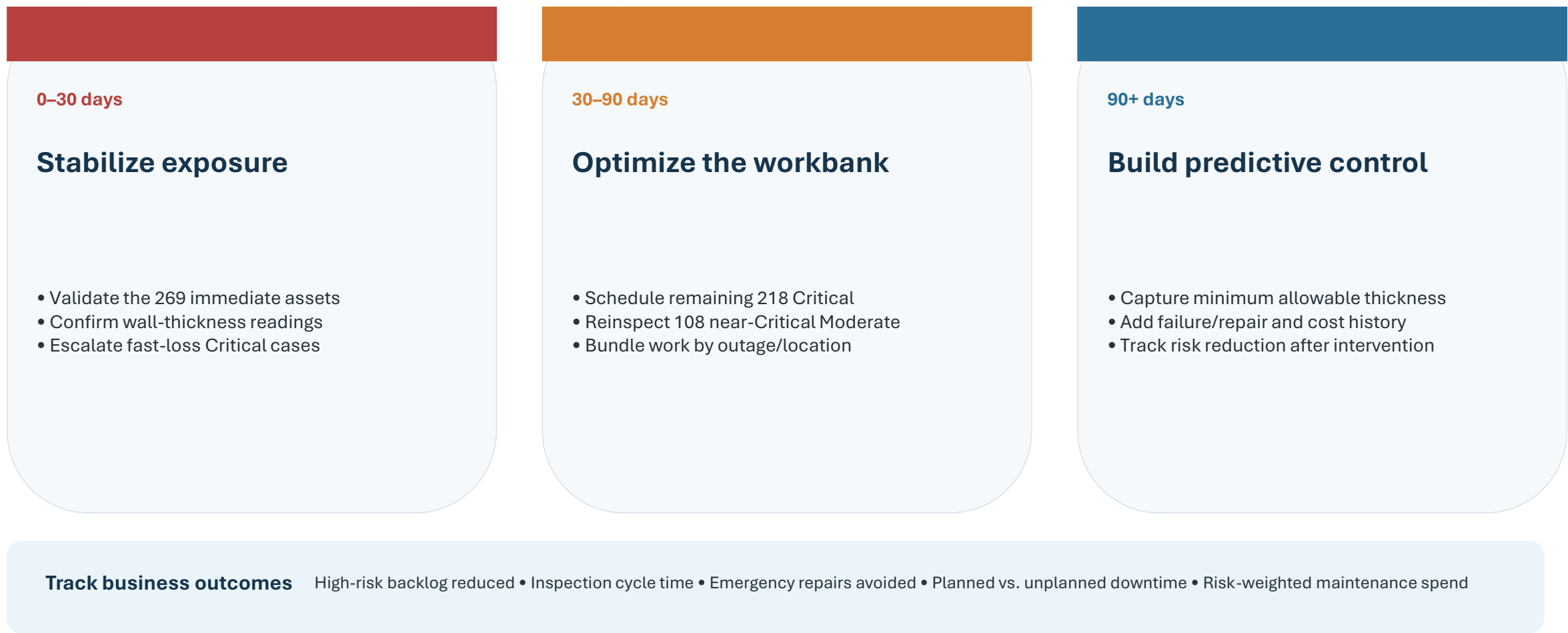
Median thickness-loss rate (mm/year)



Business implication: dynamic intervals can prevent surprise escalation while reducing unnecessary inspections on stable assets.

Convert the analysis into an integrity-management operating model

The dataset supports prioritization; additional fields are needed for defensible failure probability and financial impact.



Do not use this model as a replacement/repair decision by itself; engineering limits and field validation remain required.